

Common Ground: Establishing Group Awareness for Cross-Virtuality Collaborative Geospatial Analysis

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ABSTRACT

Virtual Reality (VR) and desktop users struggle to coordinate due to scale differences when analyzing geospatial data. This work explores multiscale workspace awareness techniques in geospatial analysis scenarios, such as scale-responsive cursors and vistas. A study ($n = 16$) compared Common Ground to current collaboration tools. The results reveal that 34% of users struggle to maintain awareness of their collaborator's focus, which drops to 9% with Common Ground; and 32% of users struggle to know precisely what features their collaborator referenced, decreasing to 15% with Common Ground. A follow-up with 12 geospatial practitioners provided insights into real-world applications of our system.

Index Terms: H.5.2 [User Interfaces]: User-centered design; H.5.3 [Group and Organization Interfaces]: Computer-supported cooperative work

1 INTRODUCTION

Multiscale data analysis is powerful but complex, especially for the coordination of multiple parties simultaneously traversing multiple datasets from differing scales and perspectives. This complexity is particularly evident in geospatial analysis, where diverse instruments collect complementary data at varying spatial scales, all of which are essential for establishing geologic context. Although unified coordinate systems (e.g., planetocentric latitude and longitude) can link collaborators, the multiscale nature of geospatial data-ranging from centimeters to kilometers per pixel-introduces complexities for maintaining spatial context. Existing collaborative geospatial analysis approaches often limit participants to a single map scale, where traditional methods like gaze sharing and annotations are typically sufficient. [2].

Cross-Virtuality Analytics (XVA) research addresses collaborative multiscale analysis challenges by connecting users across the reality-virtuality spectrum to blend different perspectives, (e.g., in-depth Virtual Reality views with broad 2D screen map views). Existing tools such as networked annotations [1] and perspective sharing [3, 4], offer partial solutions, but do not suffice for multiscale collaborators. For instance, a screen user's view of an in-situ VR collaborator's gaze only reveals a rough latitude and longitude at best; it does not reveal exactly what feature the collaborator is focusing on nor the precision of their data selection when highlighting or pinning it to coordinate actions for further analysis.

To address these multiscale challenges, we develop Common Ground, a system designed to provide cross-virtual collabora-

tors with shared understanding through responsive scale-adaptive workspace awareness cues. Dynamic workspace awareness is at the core of our approach, as lack of scale and precision understanding hinders task coordination and consensus.

2 CROSS-VIRTUAL PERSPECTIVES AND AWARENESS CUES

The Common Ground system leverages a unified longitude/latitude coordinate system and consists of two components: (i) Responsive Scale-adaptive Awareness Cues for multiscale coordination and (ii) Anchored Awareness Cues for shared context.

2.0.1 Responsive Scale-Adaptive Awareness Cues

An immersive vista and mouse cursor cue ensures users maintain awareness of each other's focus areas and field of view. The cue's radius grows and shrinks in response to the user's map scale, providing collaborators with an understanding of each other's workspace scale and interaction precision.

Screen View of Immersive User's Vista: The immersive user's pose—location and direction—is visualized as a dynamic vista on the 2D map. It appears as a transparent circle around the immersive user's position, expanding or contracting based on their current view frustum. This helps the screen user understand the immersive user's visible area, avoiding references outside of their view extent due to disparate map scale or natural view obstructions. Moreover, the vista responds to altitude changes, (e.g. the radius expands as an immersive user flies upward, increasing their field of view).

Immersive View of Screen User's Cursor Beam: The screen user's mouse cursor is rendered in the immersive environment as a vertical beam intersecting the terrain. The beam scales in height and width proportionally to the screen user's zoom level (i.e., map scale), unveiling the uncertainty in precision of their cursor position within the context of the immersive environment's scale. This reduces uncertainty for the immersive user when interpreting the screen user's selections. For instance, the screen user's mouse beam may span multiple features, and give a data readout that corresponds to a region rather than a single geologic feature. The mouse cursor visual lets the immersive user know that they need clarification on the selection or ask the screen user to zoom in to increase precision. If the screen user's zoom is limited, the immersive user can visually analyze the feature in question and precisely place an anchored annotation for the screen user to read.

2.0.2 Traditional Anchored Awareness Cues

For precise references, Common Ground implements brushing and linking for pin and shape annotations. If a screen user brushes a region (shape) on a 2D map, the corresponding area in the immersive view is also highlighted by the linked shape, and vice versa. This allows users to focus on specific subsets of data across the 2D and 3D representations while preserving their scale and spatial context.

2.1 Common Ground Multi-layered System Design

Calibrated Data Layer: This layer aligns data and interactions between GIS and immersive environments, ensuring geographic con-

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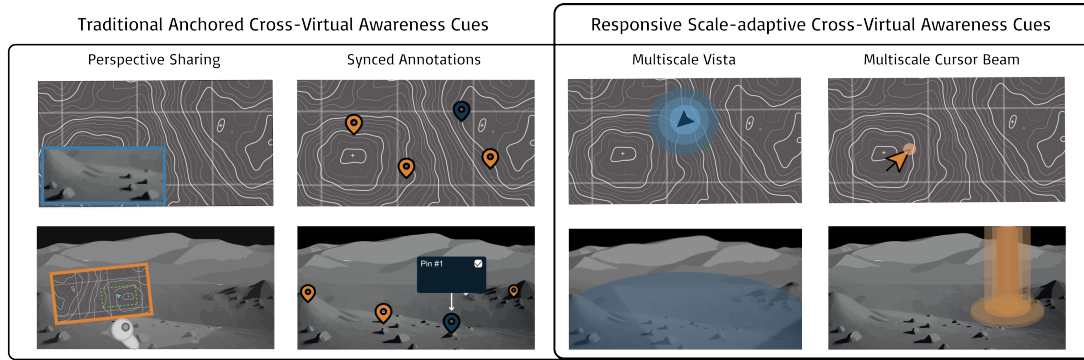


Figure 1: For screen-based users, Common Ground integrates as an overlay on traditional GIS software (top). For VR users, it connects through our game engine library to provide an immersive experience (bottom). Traditional cross-virtuality interactions are shown on the left: Perspective Sharing allows users to view another’s screen or gaze, while Annotations help highlight specific points. On the right, dynamically scaled Vista and Cursor Beam provide dynamic awareness of collaborator workspace scale and spatial precision, without sacrificing perspective.

sistency across interfaces. It handles coordinate transformations, projections, and mismatches in coordinate systems (e.g., planetographic vs. planetocentric in our Implementation with JMARS and Cesium GIS providers).

Screen GIS Integration: Common Ground overlays Unity onto GIS applications (e.g., Google Maps, JMARS, QGIS). It configures the map view’s screen-space position and dimensions, enabling synchronization of overlays and tools. Mouse teleportation allows seamless GIS control from within the Common Ground interface.

Awareness Cue Layers: These layers interface with the calibrated data layer to calculate latitude longitude coordinates of the cues (e.g., poses, mouse, vista, pins, and shapes). They leverage Unity’s Netcode for GameObjects to share cues and render them for their respective platforms (Screen-based or VR).

3 EVALUATION AND RESULTS

We evaluated Common Ground through two IRB-approved user studies comparing its multiscale visualization tools to traditional methods (screen sharing and anchored annotations). The first study involved 12 geospatial practitioners, while the second expanded to 16 participants with diverse backgrounds and simplified tasks spanning collaborative navigation and task coordination, making the study more broadly applicable to cross-virtual, multiscale collaboration.

Study 1: Participants included planetary geology experts who completed realistic geologic analysis tasks. Qualitative thematic analysis of audio transcripts revealed that Common Ground improved efficiency in communication and decision-making. As one participant noted, “This tool should help bridge gaps in geological mapping, allowing multiple people to view and resolve data together.” Experts emphasized its potential in mission-critical applications, such as Mars Rover operations.

Study 2: Participants engaged in navigation, path-finding, and coordination tasks. Awareness Questionnaire results as seen in Figure 2 show that without Common Ground, 34% of users struggled to understand their collaborator’s focus, dropping to 9% with Common Ground. Similarly, difficulty in understanding references decreased from 32% to 15%, and issues with directional guidance dropped from 25% to 13%.

Across both studies, Common Ground’s awareness cues enhanced collaboration by improving focus, reducing ambiguity, and supporting seamless communication across scales and platforms.

4 CONCLUSION

In this work we explored the efficacy of Common Ground, our system for establishing shared context between cross-virtual multiscale collaborators. We presented scale adaptive awareness cues—mouse

beam and vista—which improved users’ understanding of their collaborators’ current focus, directions, and references to objects in the scene compared to the use of the traditional tools alone.

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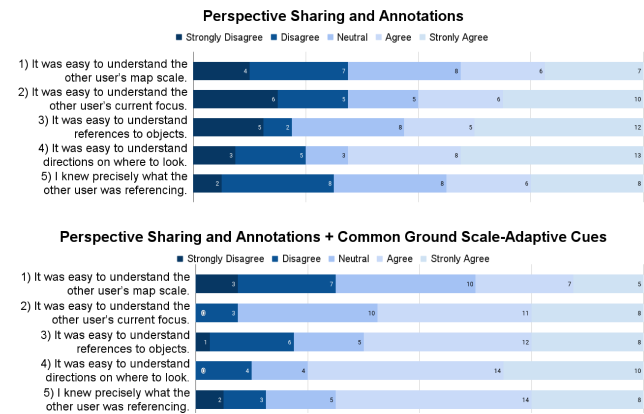


Figure 2: Group Awareness Questionnaire.

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